MIL-C-46228A(MU)
6 July 1973
SUPERSEDING
MIL-C-46228(MU)
31 May 1960

# MILITARY SPECIFICATION

# CARTRIDGE, DELAY: M93 ASSEMBLY

## 1. SCOPE

1.1 This specification covers Cartridge, Delay: M93, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

*SPECIFICATIONS*

Military

| | |
|---|---|
| MIL-A-2550 | - Ammunition and Special Weapons, General Specification for |
| MIL-E-5272 | - Environmental Testing, Aeronautical and Associated Equipment, General Specification For |
| MIL-C-60539 | - Coating, Anodic, Conventional for Aluminum and Aluminum Alloys |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | - Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-109 | - Quality Assurance Terms and Definitions |
| MIL-STD-413 | - Visual Inspection Guide for Rubber O-Rings |
| MS-28741 | - Hose Assembly, Detachable End-Fitting, Medium Pressure |

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

2.1 (Cont'd)

DRAWINGS

Munitions Command

| | |
|---|---|
| D8593312 | - Cartridge, Delay; M93, Assembly |
| B8799621 | - Carton, Packing, Ammunition For Cartridge, Delay, M93, Assembly |
| B8799620 | - Box, Packing, Ammunition For Cartridge, Delay, M93 Assembly |
| IEL 8593312 | - Inspection Equipment List for Cartridge, Delay: M93 Assembly |

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 Surface Texture

(Application for copies should be addressed to the American National Standards Institute, (ANSI), 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of delay charge, igniter and propellant charge.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.-The cartridge when tested in an M31 or M32A1 Initiator, shall produce the pressure and delay times at the temperatures specified in Table I.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

TABLE I

| Temperature | Pressure (psi) 1/ Minimum | Pressure (psi) 1/ Maximum | Delay Time, Seconds (see 6.2) |
|---|---|---|---|
| -65 to -70 F | 500 | 1250 | 1.3 max. |
| 65 to 75 F | 600 | 1500 | 1.0 ± .2 |
| 160 to 165 F | 700 | 1700 | .8 min. |

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of hose (Size No. 4) in accordance with MS 28741.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridges for primer sensitivity, 5 cartridges for sealant tests (to be selected from primer sensitivity sample), 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample) and 10 sets of unassembled components for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source,the sample shall be inspected for all requirements of the drawing and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE II

| Temperature | Vibrated | Not Vibrated |
|---|---|---|
| -65° to -70°F | 4 | 6 |
| 65° to 75°F | 4 | 6 |
| 160° to 165°F | 4 | 6 |

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-An ammunition lot of cartridges shall consist of an identifiable group of cartridge assemblies consisting of:

a. Components produced by a homogeneous process.
b Delay elements from not more than one lot.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.3.1.2 (Cont'd)

c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Propellant from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor class, the criteria in Table IV shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber "O" rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

| | | Major | | Minor | |
|---|---|---|---|---|---|
| Lot Size | Sample Size | Accept | Reject | Accept | Reject |
| 1-300 | 100% examination | - | - | - | - |
| 301-1300 | 110 | 1 | 2 | 2 | 3 |
| 1301-3200 | 150 | 2 | 3 | 3 | 4 |
| 3201-8000 | 225 | 3 | 4 | 4 | 5 |
| 8001-22,000 | 300 | 4 | 5 | 5 | 6 |

TABLE IV

| | Total Allowable Defects Per Sample Size | | | |
|---|---|---|---|---|
| Sample Size for Major Defects Number of Defects in Class | 110 | 150 | 225 | 300 |
| 2-3 | 1 | 2 | 3 | 4 |
| 4-7 | 2 | 3 | 4 | 5 |
| 8-19 | 3 | 4 | 5 | 7 |
| 20-43 | 4 | 5 | 8 | 10 |
| Over 43 | 6 | 8 | 11 | 14 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

| Sample Size for Major Defects<br>Number of Defects in Class | Total Allowable Defects Per Sample Size<br>110 | 150 | 225 | 300 |
|---|---|---|---|---|
| 2-3 | 2 | 3 | 4 | 5 |
| 4-7 | 3 | 4 | 5 | 7 |
| 8-16 | 4 | 5 | 8 | 10 |
| 17-48 | 6 | 8 | 11 | 14 |
| over 48 | 8 | 11 | 17 | 20 |

4.3.2.1 Sealing disc, Drawing C8594077, covering a detail of Drawing D8593312.

| Defect | Method of Inspection |
|---|---|
| CRITICAL | |
| 1. Metal defective | Visual |
| 2. Marking missing, incorrect or illegible | Visual |
| MAJOR | |
| 101. Diameter, min. | Gage |
| 102. Thickness | Gage |
| 103. Protective finish inadequate or defective | Visual |
| 104. Burr | Visual |
| MINOR | |
| 201. Diameter, max. | Gage |
| 202. Varnish missing or inadequate | Visual |

4.3.2.2 Case, Cartridge, Drawing C8595202 covering a detail of Dwg. D8593312.

| | |
|---|---|
| CRITICAL | |
| 1. Metal defective | Visual |
| MAJOR | |
| 101. Concentricity of case head with body | Gage |
| 102. Straightness of base | Gage |
| 103. Thickness of base | Gage |
| 104. Thickness of wall | Gage |
| 105. Protective finish inadequate or defective | Visual |
| 106. Burr or sharp corner | Visual |
| 107. Foreign matter | Visual |
| 108. Diameter of head, min. | Gage |
| 109. Hardness | Gage |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.3.2.2 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MINOR** | |
| 201. Diameter of head, max. | Gage |
| 202. Diameter of body, max. | Gage |
| 203. Total length | Gage |
| 204. Location of datum diameter | Gage |
| 205. Thickness of wall, max. | Gage |
| 206. Thickness of base, max. | Gage |
| 207. Radius missing or incorrect | Visual |
| 208. Surface finish improper | Visual |

4.3.2.3 Delay Element, M8, Dwg. C8593163 covering a detail of Dwg. D8593312.

| | |
|---|---|
| **CRITICAL** | |
| 1. Metal defective | Visual |
| 2. Mixed delay elements | Visual |
| 3. Flash hole missing or obstructed | Visual |
| 4. Marking missing, incorrect or illegible | Visual |
| 5. Delay charge not visible through flash hole | Visual |
| 6. Primer uncrimped or seal missing | Visual |
| **MAJOR** | |
| 101. Protective finish inadequate or defective | Visual |
| 102. Foreign matter | Visual |
| 103. Burr | Visual |
| **MINOR** | |
| None defined | |

4.3.2.4 Cartridge, Delay, M93, Assembly, Dwg. D8593312.

| | |
|---|---|
| **CRITICAL** | |
| 1. Any part missing | X-ray 1/ and Visual |
| 2. Improper assembly | X-ray 1/ and Visual |
| 3. Mixed delay elements or propellant | X-ray 1/ and Visual |
| 4. Metal defective | Visual |
| 5. Marking missing, incorrect or illegible | Visual |
| 6. Crimp missing, incorrect or illegible | Visual |



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.3.2.4 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| **MAJOR** | |
| 101. Cartridge profile failure | Gage |
| 102. Diameter of body, max. | Gage |
| 103. Diameter of head, max. | Gage |
| 104. Location of datum diameter | Gage |
| 105. "O" Ring split, cracked or cut | Visual |
| 106. Protective finish inadequate or defective | Visual |
| 107. Foreign matter | Visual |
| 108. Burr | Visual |
| **MINOR** | |
| 201. Total length, max. | Gage |
| 202. Varnish missing or inadequate | Visual |
| 203. Case damaged | |

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Packing, Ammunition for Cartridge Delay, M93, Drawing B8799621.

CRITICAL

None defined

MAJOR

None defined

MINOR

| | |
|---|---|
| 201. Contents loose | Manual and Visual |
| 202. Quantity of contents incorrect | Visual |
| 203. Separator or filler missing | Visual |
| 204. Cut or damaged through impregnated box | Visual |
| 205. Marking missing, incorrect or illegible | Visual |

4.3.2.6 Box, Packing, Ammunition for Cartridge Delay, M93, Drawing B87699620.

CRITICAL

None defined



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

| Defect | Method of Inspection |
|---|---|
| MAJOR | |
| None defined | |
| MINOR | |
| 201. Contents loose | Manual and Visual |
| 202. Hardware missing, loose, broken or improperly engaged | Visual |
| 203. Strapping missing, loose, broken or improperly assembled | Visual |
| 204. Contents of box exposed | Visual |
| 205. Box damaged | Visual |
| 206. Marking missing, incorrect or illegible | Visual |
| 207. Number of cartons incorrect | Visual |
| 208. Car seal missing, unsealed or improperly positioned | Visual |

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105 and MIL-C-60539.

4.3.3.2 Vibration.-Twelve of the 30 assembled x-rayed cartridges of the initial production sample shall be subjected to vibration in accordance with 3.3. After vibration the 12 vibrated cartridges shall be x-rayed as directed in note 1/ 4.3.2.4. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the cartridge or in the lot of primers, 12 additional cartridges shall be selected and tested as indicated herein.

4.3.3.3 Airtightness.-Each cartridge shall be subjected to an airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of one cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.3.3.5 Sealant application.-A sample of 5 cartridges (consumed in the primer sensitivity test) shall be disassembled and examined for improper sealant application. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.7 Functional.-The following tests shall be performed by the government.

4.3.3.7.1 Ballistics.-Thirty test cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.7.1.1 and Table V. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

Table V

| Temperature | Sample Size |
|---|---|
| -65° to -70°F | 10 |
| 65° to 75°F | 10 |
| 160° to 165°F | 10 |

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8593312.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge cases shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Airtightness.-The method and procedure for performing the airtightness test shall be by use of equipment listed on IEL 8593312.

4.4.4 Primer sensitivity.-Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery, one inch, maximum, above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inches in diameter, wieghing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the procuring activity.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been properly



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

4.4.5 (Cont'd)

applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Hardness.-A hardness testing machine shall be used to determine the hardness values specified on the applicable drawing.

4.4.7 Functional.

4.4.7.1 Ballistics.-The sample for ballistic tests shall be grouped and temperature conditioned in accordance with 4.4.7.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.4.7.1.1.

4.4.7.1.1 Temperature conditioning.-A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawing B8799621 and Drawing B8799620.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class C
DOT Marking - Explosive Power Devices - Class C
Handle Carefully - (Keep Fire Away)



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU(FA)

Project No. 1377- A530



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-46228A(MU) Cartridge, Delay: M93 Assembly

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

U.S. MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SMUFA-J4000
Philadelphia, Penna 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T05:17Z
Check the source to verify that this is the current version before use.